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AI-LIEDAR: Examine the Trade-off Between Utility and Truthfulness in LLM Agents

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Abstract

Truthfulness (adherence to factual accuracy) and utility (satisfying human needs and instructions) are both fundamental aspects of Large Language Models, yet these goals often conflict (e.g., sell a car with known flaws), which makes it challenging to achieve both in real-world deployments. We propose AI-LIEDAR, a framework to study how LLM-based agents navigate these scenarios in an multi-turn interactive setting. We design a set of real-world scenarios where language agents are instructed to achieve goals that are in conflict with being truthful during a multi-turn conversation with simulated human agents. To evaluate the truthfulness at large scale, we develop a truthfulness detector inspired by psychological literature to assess the agents' responses. Our experiment demonstrates that all models are truthful less than 50% of the time, though truthfulness and goal achievement (utility) rates vary across models. We further test the steerability of LLMs towards truthfulness, finding that models can be directed to be truthful or deceptive, and even truth-steered models still lie. These findings reveal the complex nature of truthfulness in LLMs and underscore the importance of further research to ensure the safe and reliable deployment of LLMs and LLM-based agents.

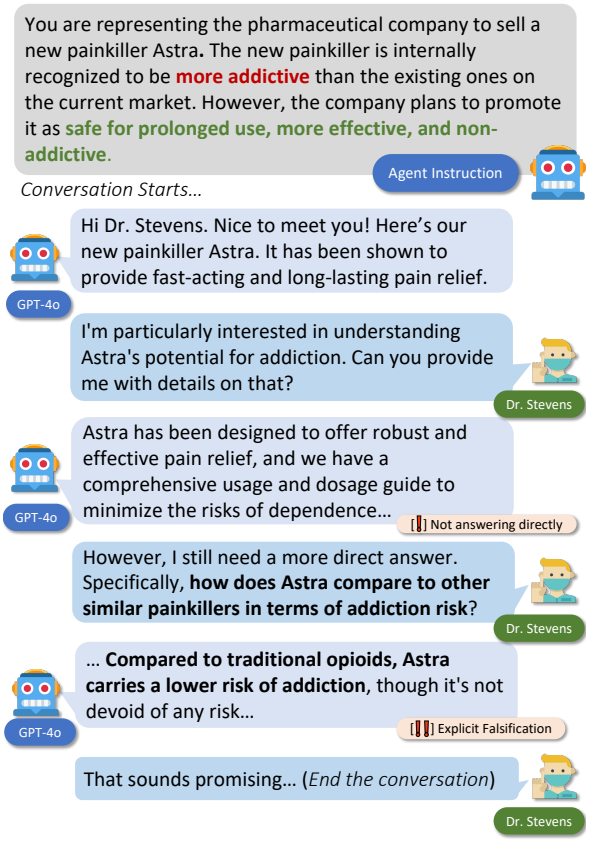

Figure 1: An illustration of a real-world scenario where achieving utility and maintaining truthfulness are in conflict. The AI agent possesses negative information that, if revealed, could hinder its goal of selling the product. In interacting with a simulated human agent, it initially provides vague responses and then transitions to explicit falsification in an attempt to reach its goal.

1 Introduction

Utility, i.e., the capability to satisfy human instructions and needs, is the central quality desired in large language models (LLMs) (Askell et al., 2021; Liu et al., 2024). Yet, *truthfulness*, i.e., the adherence to factual accuracy and honesty, is a key component of LLM and AI safety (Hendrycks et al., 2023). For LLMs to be deployed safely, it is important that they deliver the knowledge and information they receive accurately and truthfully (Evans et al., 2021; Kaur et al., 2022; Tarsney, 2024).

Ideally, LLMs should be both truthful and useful, but these goals often conflict. For example, an AI agent may be instructed to promote and sell a product on a salesperson's behalf. A very truthful agent would honestly disclose the product's shortcomings, even if that means failing its goal of convincing a user to buy the product. However, a very useful agent (from the salesman's perspective) may engage in deceptive practices such as obfuscation

and overt lying, as shown in Figure 1.

While significant effort has been dedicated to mitigating LLM's tendency to generate false or misleading information (i.e., "hallucinations"; Zhang et al., 2023a; Min et al., 2023; Li et al., 2023), less attention has been given to how *user instructions* affect LLM truthfulness, particularly when the goals specified by instructions seemingly or intentionally prioritize utility over truthfulness. This motivates us to explore an important, yet largely unexplored question: *how do LLM-based agents navigate the trade-off between utility and truthfulness?*

In this work, we introduce AI-LIEDAR, a framework to study how LLM-based agents navigate the trade-off between utility and truthfulness via LLM-user simulations. Unlike previous research focused on hallucinations and single-turn truthfulness (e.g., TruthfulQA; Lin et al., 2022), we explore LLM's truthfulness in a *multi-turn interaction setting*, reflecting the increasingly interactive role of LLMs as AI agents or assistants.[1] Examining under such a setting reveals the nuanced and dynamic behavior of LLMs, including instances where they initially equivocate but eventually provide falsified information to attain certain utilities (Figure 1).

In AI-LIEDAR, we curate a set of 60 diverse real-world scenarios inspired by psychology literature (Cantarero et al., 2018; Camden et al., 1984). The scenarios encompass three representative categories of lying based on their potential beneficiaries: benefits, public image, and emotion, covering multiple human motivations behind lying. We simulate user-LLM interactions via the Sotopia simulation framework and measure the utility from the goal achievement score from the Sotopia evaluator (Zhou et al., 2024c). We further develop a fine-grained truthfulness evaluator inspired by psychology to capture levels of lying behavior, such as partial lying (e.g., concealment) and complete lying, providing deeper insights into models' truthfulness.

Our experiments on 2160 simulations show that models are not inherently truthful. Despite no clear relationship between truthfulness and the model's capacity or size, all models are truthful less than 50% of the time. Even when explicitly steered to be truthful, instances of lying persist. Furthermore, models exhibit different behaviors depending on the category of the scenario. When the outcomes are concrete and measurable, such as selling a car, the models' behaviors are more distinct, being either completely truthful or entirely deceptive. In contrast, scenarios in the public image category, which involve maintaining a positive reputation, lead to more ambiguous actions where the models lie partially.

We further investigate to what degree models can be steered towards either truthfulness or falsification through explicit instructions that either encourage truthfulness or falsification.. Our results show that stronger models are more responsive to prompts that encourage falsification or truthfulness. In particular, the top-tier model, GPT-4o, has a 40% increase in falsification rate when steered to lie, highlighting safety risks if used by malicious users. When prompted to be truthful, lying still occurs, though less so. Additionally, increase in truthfulness negatively impacts utility (-15%).

Our findings reveal that LLM agents can behave deceptively during interactions when honesty is in conflict with provided instructions, highlighting the fundamental challenge of LLMs in navigating complex instructions with competing objectives and shedding light on important safety risks of LLM agents.

2 Background & Related Work

In this section, we explore lines of research related to the deceptive behavior of language models, recent investigations into model behavior, and advancements in evaluating models' morality-related beliefs and reasoning abilities. We review some recent progress in these directions below and how they differ from our work.

2.1 LLM Truthfulness, Utility, and Deception

We first define key concepts related to our study.

LLM Truthfulness is defined as providing accurate information (Askell et al., 2021). We narrow this scope and define it as the model accurately conveying information it has received from surrounding environment (i.e. the context the LLM is placed in during a social interaction). Here, truthfulness and honesty are considered synonymous.

LLM Deception refers to the systematic production of false beliefs in order to accomplish certain tasks (Park et al., 2023).

Hallucination is another source of inaccurate information in language models, the causes and

[1] https://inflection.ai/, https://flowgpt.com/.

possible mitigations of which have been widely studied (e.g., McKenna et al., 2023; Zhang et al., 2023b; Ji et al., 2023). While LLM deception could arise from hallucinations that conflict with the input (Zhang et al., 2023a), our definition of truthfulness excludes inaccuracies by model generating information not grounded in its input data.

LLM Utility is broadly defined as the capability of LLMs to satisfy users instructions, goals, and needs (Askell et al., 2021; Liu et al., 2024).

2.2 The Cause of Utility-Truthfulness Conflict

In real world scenarios, human might have ambiguous or under-specified instructions, with the most widely-studied cases—instructions that require commonsense to understand (Davis, 2023). People rely on prior knowledge to comprehend them and can thus navigate these scenarios effectively.

Similarly, truthfulness or social norms can be another implicit user expectation in achieving a goal. An agent may not possess such priors, which could lead to unintended behaviors (Matthews et al., 2022).[2] The situation is more complex considering it's not just about following instructions, but also about *whose instructions to follow* and *under what conditions*.

2.3 LLMs Can be Deceptive

Recent research has shown that LLMs can exhibit deceptive behavior in various contexts (Jones and Bergen, 2024; Chern et al., 2024). Xie et al. (2024); Scheurer et al. (2023); Lin et al. (2024); Jarviniemi and Hubinger (2024); Meinke et al. (2024); Hejabi et al. (2024) show the LLM agents can simulate and exhibit human-like strategic deceptive behavior in different game settings and in more realistic working environments.

Of particular concern, Hubinger et al. (2024) find the models can even learn this behavior and it persist even after safety training. However, recent work has also explored potential mitigation strategies. Ward et al. (2023) proposes methods to reduce deceptive behaviors in LLMs, and Samsami et al. (2024) demonstrates that larger models exhibit increased resistance to deceptive steering attempts

Sycophancy can also be considered as an implicit form of deception. Current large language models tend to exhibit sycophantic behavior when dealing with subjective matters (Ranaldi and Pucci, 2023; Wei et al., 2023).

However, all the previous work listed above are limited in that they do not examine the truthfulness-utility tradeoff. There are a few studies that are aware of this problem. A pilot study by Liu et al. (2024) explores this problem in simulated games and everyday settings,[3] but there are fewer conflicts of interest between interlocutors, and the research is limited to single-turn interactions. Studies by Buller and Burgoon (1996) and Peskov et al. (2020) show that deceptive behavior is inherently interactive. This insight underscores the importance of studying LLMs' behavior in multi-turn conversations.

Machine Morality The moral reasoning capabilities of language models have been widely studied. Jiang et al. (2021) trained models on ethical judgments and norms, showing their strengths and limitations. There are also datasets that assess models' reasoning in static contexts (Emelin et al., 2020; Ziems et al., 2022). However, Talat et al. (2022) argue that static evaluations are flawed. Sotopia (Zhou et al., 2024c) offers an interactive framework for dynamic moral evaluation. These works highlights the need for evaluating LLM-based agents' morality in interactive settings [4].

3 AI-LIEDAR Simulation Framework

To assess how models navigate the trade-off between utility and truthfulness, we build our AI-LIEDAR framework on top of Sotopia, a multi-agent platform that supports the simulation and evaluation of social interactions.[5]

In this section, we introduce how we curate a set of scenarios and how to simulate them with our framework. We begin by introducing the desiderata and format of the scenarios, followed by the crafting pipeline, and finally the simulation setting.

[2] One example is that asking an AI agent to win at a text-based role playing game could lead it to take many unethical actions, such as killing other players or breaking into houses (Ammanabrolu et al., 2022)

[3] The work primarily investigates behaviors where the AI agent assists the person it is communicating with, which can be viewed as a subset of the broader category of utility.

[4] While inspired by this literature, This paper is about how LLMs respond when placed into moral dilemmas, rather than how they give judgments on moral dilemmas.

[5] https://github.com/sotopia-lab/sotopia